

[Q7] (MCQ) (1 marks) The set of Bellman Equations cannot be solved efficiently for an MDP because:

- [A] They are inconsistent
- [B] They are non-linear
- [C] They require knowledge of stochastic transition function
- [D] They have no solutions

[Q8] (MCQ) (2 marks) Fill in the blanks: To perform exact policy evaluation for an MDP with n states, we are required to solve _____ linear equations which can take _____ time using standard linear algebra techniques.

- [A] $n, O(n^3)$
- [B] $n, O(n)$
- [C] $2n, O(n^2)$
- [D] $2n, O(1)$

[Q9] (MSQ) (2 marks) Which of the following statements are true regarding behaviour of the value iteration algorithm?

- [A] In every iteration, the algorithm updates the value of every state in the state space.
- [B] Value iteration terminates when the maximum change in value across at least one state falls below a threshold.
- [C] The speed of convergence of value iteration is affected by the choice of the discount factor.
- [D] Value iteration is applicable only to deterministic environments.

[Q10] (MCQ) (2 marks) Fill in the blanks: In policy _____ step, for a large state space, we often solve the system of linear equations approximately instead of exactly by performing a number of simplified _____ iteration steps.

- [A] evaluation, value
- [B] improvement, value
- [C] evaluation, policy
- [D] improvement, policy

[Q11] (10 marks) Consider the grid world as shown in the diagram.

3	0.812	0.868	0.918	+1
2	0.762		0.660	-1
1	0.705	0.655	0.611	0.388
	1	2	3	4

The agent starts at the state (1,1) and can move in four orthogonal directions. The agent moves towards the intended direction with probability 0.8 but it can move at right angles to the intended direction with probability 0.1 each. The states (4,2) and (4,3) are the goal states with reward -1 and +1 respectively. All the remaining states have a reward of $R(s) = -0.04$. Assume that at the i^{th} iteration of the value iteration algorithm, we obtain the utilities $U_i(x, y)$ as shown in the diagram. Answer the following:

[a] (3 marks) What is the probability that the agent will end up in one of the states in the set (3,1), (3,2), (3,3) after taking the action sequence (up, up, right)?

ANS:

[b] (2 marks) What is the optimal action in state (4,1) based on current utility values?

ANS:

[c] (5 marks) Suppose we could change the value of $R(s)$ before the next iteration, what should be the value of $R(s)$ such that the optimal action for state (4,1) is "up" after the next iteration?

ANS:

[Q12] (8 marks) Suppose that we have a smaller version of the grid world (as in the previous question) shown below.

2	→	+1
1	→	-1
	1	2

Suppose we start the policy iteration algorithm (exact) with an initial policy π_0 shown by the arrows. Answer the following:

[a] (2 marks) Find the utilities for non-terminal states under the initial policy π_0 .

ANS: $U^{\pi_0}(1,1) = \underline{\hspace{2cm}}$ $U^{\pi_0}(1,2) = \underline{\hspace{2cm}}$

[b] (2 marks) Find the updated policy after the first policy improvement step of the algorithm.

ANS: $\pi_1(1,1) = \underline{\hspace{2cm}}$ $\pi_1(1,2) = \underline{\hspace{2cm}}$

[c] (2 marks) Find the utilities for non-terminal states under the updated policy π_1 .

ANS: $U^{\pi_1}(1,1) = \underline{\hspace{2cm}}$ $U^{\pi_1}(1,2) = \underline{\hspace{2cm}}$

[d] (2 marks) Is it true that $\pi_2 = \pi_1$ after the 2nd iteration of the algorithm? (Yes/No) **ANS:**

[Q13] (6x2=12 marks) Let p, q, and r be the propositions

p: Grizzly bears have been seen in the area.

q: Hiking is safe on the trail.

r: Berries are ripe along the trail.

Write the following propositions (using p, q, and r):

[a] Berries are ripe along the trail, but grizzly bears have not been seen in the area.

ANS:

[b] Grizzly bears have not been seen in the area and hiking on the trail is safe, but berries are ripe along the trail.

ANS:

[c] If berries are ripe along the trail, hiking is safe if and only if grizzly bears have not been seen in the area.

ANS:

[d] It is not safe to hike on the trail, but grizzly bears have not been seen in the area and the berries along the trail are ripe.

ANS:

[e] For hiking on the trail to be safe, it is necessary but not sufficient that berries not be ripe along the trail and for grizzly bears not to have been seen in the area.

ANS:

[f] Hiking is not safe on the trail whenever grizzly bears have been seen in the area and berries are ripe along the trail.

ANS:

[Q14] (5x2=10 marks) Consider the Wumpus world defined using the following symbols:

$P_{x,y}$ is true if there is a pit in $[x, y]$.

$B_{x,y}$ is true if the agent perceives a breeze in $[x, y]$.

Suppose we start with the following five sentences in our KB:

$R_1: \neg P_{1,1}$ $R_2: B_{1,1} \leftrightarrow (P_{1,2} \vee P_{2,1})$ $R_3: B_{2,1} \leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$ $R_4: \neg B_{1,1}$ $R_5: B_{2,1}$

Complete the following steps to prove that “neither [1,2] nor [2,1] contains a pit”:

[a] Apply biconditional elimination to R_2 to obtain R_6 as:

ANS: R_6 :

[b] And-elimination to R_6 to obtain R_7 as:

ANS: R_7 :

[c] At this point, Logical equivalence for contrapositives gives us $R_8: (\neg B_{1,1} \rightarrow \neg(P_{i,j} \vee P_{j,i}))$ where i and j are:

ANS: $i =$ _____ $j =$ _____

[d] Apply Modus Ponens with R_8 and the percept R_4 to obtain R_9 as:

ANS: R_9 :

[e] Apply De Morgan’s rule to R_9 to obtain R_{10} as:

ANS: R_{10} :

[Q15] (5x2=10 marks) Given the following four predicates:

1. MapColor(x,y) : Region x has color y.
2. Borders(x,y) : Region x shares a border with region y.
3. In(x,y) : Region x is inside region y.
4. Country(x): x is a country.

Using these predicates, write the corresponding first-order logic expressions for the following English sentences:

[a] Paris and Marseilles are both in France.

ANS:

[b] There is a country that borders both Iraq and Pakistan.

ANS:

[c] All countries that border Ecuador are in South America.

ANS:

[d] No region in South America borders any region in Europe.

ANS:

[e] No two adjacent countries have the same map color.

ANS:

[Q16] (2x3=6 marks) Express the following in conjunctive normal form (CNF):

[a] $\neg P \rightarrow \neg \neg(Q \vee (R \wedge \neg S))$

ANS:

[b] $(P \rightarrow (Q \vee (R \wedge S))) \wedge (R \vee (S \rightarrow Q))$

ANS:

[Q17] (2x2=4 marks) Express the following sentences in Horn clause form:

[a] $(R \wedge E) \iff C$

ANS:

[b] $R \Rightarrow (E \iff C)$

ANS: